ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Measurement of Length and Motion** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** metre	Metre is the unit of length.
2	**(c)** linear motion	Straight path.
3	**(d)** above the mark	Straight above.
4	**(a)** 100	1 m is 100 cm.
5	**(a)** rotational motion	Fan rotates.
6	**(b)** one end of the object	Start at zero.
7	**(d)** 1000	1 km is 1000 m.
8	**(c)** periodic motion	To and fro.
9	**(c)** a curved line	Thread follows curves.
10	**(d)** 10	1 cm is 10 mm.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Metre (1); Symbol m (1).	1+1
12	Ruler (1); Measuring tape (1).	1+1
13	Hands of a clock (1); A stone whirled on a string (1).	1+1
14	3 x 100 (1); 300 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Metre (1); Centimetre (1); Kilometre (1).	1+1+1
16	Linear (1); Circular (1); Periodic (1).	1+1+1
17	Take a thread (1); Lay along the line (1); Measure thread on a scale (1).	1+1+1
	OR: 1 m = 100 cm (1); 1 cm = 10 mm (1); 1 km = 1000 m (1).	1+1+1
18	Car (1); Train (1); Falling apple (1).	1+1+1
	OR: Book: cm (1); Cities: km (1); Coin: mm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Place the ruler along it (1); Zero at one end (1); Eye above the other end (1); Read the mark (1); Write with unit (1).	1+1+1+1+1
	OR: Zero at start (1); Straight along (1); Eye above (1); Read accurately (1); Write unit (1).	1+1+1+1+1
20	2 km = 2000000 mm (1); 3 m = 3000 mm (1); 4 cm = 40 mm (1); Using multiplication (1); With units (1).	1+1+1+1+1
	OR: Motion: change of position (1); Moving car (1); Rest: no change (1); House (1); Depends on observer (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up conversions: 1 m = 100 cm (not 10 cm), 1 cm = 10 mm, 1 km = 1000 m.
- Reading the scale from the side, which gives a wrong value; keep the eye straight above the mark.
- Starting the measurement from the 1 cm mark without subtracting; either start at zero or subtract the first reading.
- Stretching or loosening the thread while measuring a curved line.
- Calling a spinning top **circular** motion; spinning about its own axis is rotational.
- Saying all repeating motion is periodic; it must repeat after **equal** intervals of time.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Measurement of Length and Motion** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** circular motion	Goes around.
2	**(d)** elbow to fingertip	Ancient unit.
3	**(a)** ruler	Ruler.
4	**(c)** random motion	Irregular.
5	**(d)** non-standard unit	Varies between people.
6	**(a)** kilometre	Kilometre.
7	**(a)** rotational motion	Spins on axis.
8	**(c)** measuring tape	Tape bends.
9	**(a)** kilogram	Kilogram measures mass.
10	**(c)** a book on a table	It does not change position.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Change of position with time (1); E.g. a running boy (1).	1+1
12	Swing (1); Pendulum (1).	1+1
13	2 x 1000 (1); 2000 m (1).	1+1
14	No change in position (1); E.g. a book on a table (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	400 cm (1); 4000 mm (1); Multiply (1).	1+1+1
16	Clock hand (1); Stone whirled on a string (1); Tip of a fan blade (1).	1+1+1
17	Standard length (1); Same for all (1); Easy to read (1).	1+1+1
	OR: Divide by 1000 (1); 7 km (1); Answer with unit (1).	1+1+1
18	Pendulum (1); Swing (1); Vibrating string (1).	1+1+1
	OR: Ruler (1); Tape (1); Metre rod (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Tailoring (1); Cooking (1); Building houses (1); Sports tracks (1); Travel distances (1).	1+1+1+1+1
	OR: Measure 100 sheets (1); Note the thickness (1); Divide by 100 (1); One sheet value (1); Write in mm (1).	1+1+1+1+1
20	Car: linear (1); Fan: rotational (1); Swing: periodic (1); Hand of a clock: circular (1); Bird: random (1).	1+1+1+1+1
	OR: Linear: along a straight line (1); Linear: car on a straight road, falling stone (1); Circular: along a circle (1); Circular: stone whirled on a string, hands of a clock (1); The path decides the type of motion (1).	1+1+1+1+1

Mistakes to avoid:

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- Reading the scale from the side, which gives a wrong value; keep the eye straight above the mark.
- Starting the measurement from the 1 cm mark without subtracting; either start at zero or subtract the first reading.
- Stretching or loosening the thread while measuring a curved line.
- Calling a spinning top **circular** motion; spinning about its own axis is rotational.
- Saying all repeating motion is periodic; it must repeat after **equal** intervals of time.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Measurement of Length and Motion** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** a unit	Unit.
2	**(a)** km	Km.
3	**(b)** straight line	Vertical.
4	**(d)** metre	Metres.
5	**(d)** mm	Mm.
6	**(c)** position	Position changes.
7	**(b)** 15 cm	Realistic.
8	**(b)** measuring tape	Tape.
9	**(d)** no change in position	No change.
10	**(b)** metres	Metres.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	To get same result for all (1); To compare measurements (1).	1+1
12	5 x 10 (1); 50 mm (1).	1+1
13	A car on a road (1); A falling stone (1).	1+1
14	An ancient unit (1); Elbow to fingertip (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Divide by 100 (1); 2.5 m (1); With unit (1).	1+1+1
16	Table (1); House (1); Book (1).	1+1+1
17	It bends (1); It fits body shapes (1); It gives lengths (1).	1+1+1
	OR: Zero at the start (1); Eye above the mark (1); Write the unit (1).	1+1+1
18	Cubit (1); Foot (1); Handspan (1).	1+1+1
	OR: Butterfly (1); Ant (1); Children playing (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Take a thread (1); Place along curve (1); Mark end points (1); Straighten the thread (1); Measure on a scale (1).	1+1+1+1+1
	OR: 1500 m (1); 0.9 m (1); 60 cm (1); Use conversion (1); Include units (1).	1+1+1+1+1
20	Body parts differ (1); Cubit varies (1); Standard gives same value (1); Easier to compare (1); Used by all (1).	1+1+1+1+1
	OR: Passenger in train (1); At rest in seat (1); Moving w.r.t. platform (1); Trees appear to move (1); Depends on observer (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up conversions: 1 m = 100 cm (not 10 cm), 1 cm = 10 mm, 1 km = 1000 m.
- Reading the scale from the side, which gives a wrong value; keep the eye straight above the mark.
- Starting the measurement from the 1 cm mark without subtracting; either start at zero or subtract the first reading.
- Stretching or loosening the thread while measuring a curved line.
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